

Viva Q&A — PS 26061 · general questions

AI-Driven Smart Energy Management for Polar Research Stations · MoES / NCPOR · one-line answers · companion to the ML question set

The three facts everything else hangs off: diesel arrives at a polar station **once a year by ship**; a generator at low load wastes fuel badly; but you cannot switch it off because its **waste heat** is the station's heating. Say those three and the rest of the viva is defence.

1 Understanding the problem

Why did you choose this problem statement?	It's labelled an AI problem, but its hard part is power systems — which is exactly what an EEE team knows.
State the problem in one sentence.	A polar station gets one fuel delivery a year and must make it last twelve months without ever going dark.
Who is the end user?	The station engineer at Bharati during winter-over, plus NCPOR's logistics planners in Goa.
Who is the stakeholder ministry?	Ministry of Earth Sciences, through the National Centre for Polar and Ocean Research.
What happens if this isn't solved?	Fuel is wasted, resupply voyages stay mandatory, and a bad winter carries a real stockout risk.
How is it handled today?	Manually, with conservative margins — run a generator continuously and keep plenty of spare fuel.
Why isn't that good enough?	It's safe but wasteful, and the margin is paid for in diesel shipped thousands of kilometres.
Which station are you modelling?	Bharati — year-round crew, modern, the best documented of the three.
How many people live there?	Roughly 50–60 in the summer campaign, down to around 20–25 over winter.
Have you spoken to NCPOR?	Not yet — and we know the five things we'd ask them on day one.
What would you ask them?	Generator ratings and fuel curves, annual fuel offloaded, crew numbers, deferrable loads, and existing heat recovery.

2 The solution

Explain your solution in one sentence.	It predicts the next 72 hours, plans the next 48, and reports how many days of fuel remain.
What is the core insight?	Generator waste heat warms the station, so electricity and heat must be optimised together.
Why can't you just switch the generator off?	You lose the waste heat and must fire the boiler — often burning more than you saved.
What are the main components?	Digital twin, forecasting, optimiser, fuel ledger, operator dashboard.
Hardware or software?	Software — it's a Software category PS, with an optional hardware validation rig later.
What goes in and what comes out?	In: meter readings, weather, crew roster. Out: an hourly schedule and a days-of-autonomy figure.
What's your tech stack?	Python, OR-Tools, LightGBM, FastAPI, React — all CPU-only and offline.
What's the hero metric?	Days of autonomy remaining before the next resupply ship.
Why not measure it in rupees?	There is no electricity price in Antarctica — survival margin is the real currency.
What are deferrable loads?	Snow melting, water heating, battery charging, laundry — things that can wait a few hours safely.
And what is never deferrable?	Life support, comms, medical, and freezers holding irreplaceable ice-core samples.
Do you have a working prototype?	Be honest about your actual state — then pivot straight to what it will do and when.

3 Novelty & competition

What is genuinely new here?	We put the thermal balance inside the same optimisation as the electrical balance; existing systems don't.
Don't energy management systems already exist?	They do — they minimise cost, reactively, on electricity alone. We minimise stockout risk, predictively, on both.
Who is the closest existing system?	Princess Elisabeth Antarctica, whose EMS uses a PLC assigning priorities reactively — no forecasting horizon.
What stops someone copying this?	Nothing technical — the moat is understanding why the obvious electricity-only answer is wrong.
Is this research or a product?	A product built on a known technique applied to a domain that hasn't had it.
Would you patent it?	We'd rather publish and have NCPOR adopt it than lock it up.
What if another SIH team has the same idea?	Likely — but the heat–electricity co-optimisation is the hard part, and that's where we'd differ.

4 Feasibility & implementation

Can this actually be built?	Yes — every component is an established technique; the work is in the domain modelling.
What is the hardest part?	Calibrating the digital twin so the thermal side is physically believable.
What hardware does the station buy?	None — it reads meters and generator panels that already exist.
Where does it run?	On a normal station PC, fully offline, no cloud and no GPU.
What if the satellite link is down?	Nothing changes — it was never in the loop.
How long to deploy for real?	Staged: validate on the twin, replay historical logs, run advisory for a season, then automate.
Why advisory mode first?	The software is ready long before the operator's trust is, and that ordering is deliberate.
What do you still need?	Real NCPOR telemetry — it changes our numbers, not our architecture.
Do you need government support?	Only for data access; the build needs nothing from anyone.
What are the dependencies?	Open-source libraries only — no licensed solver, no paid API, no vendor lock-in.

5 Cost, impact & sustainability

What does it cost to build?	Effectively only our time — the entire stack is open source and runs on existing hardware.
What does it cost to run?	Negligible — one CPU process on a machine the station already powers.
What's the business model?	It's a public good — the return is fuel not burned and voyages not made, not revenue.
Then how do you justify it?	Mawson cut diesel from 600,000 to under 200,000 litres a year and moved to refuelling every second year.
What's the environmental impact?	Less CO ₂ and less black carbon — soot darkens snow, lowers albedo and accelerates melt.
Which SDGs does it touch?	SDG 7 affordable clean energy, SDG 13 climate action, SDG 9 resilient infrastructure.
How do you measure success?	Litres saved per year and days of autonomy gained against the current operating baseline.
Is it sustainable long-term?	Yes — open source, no licence costs, and retrainable by NCPOR without us.
Any policy relevance?	It supports India's Antarctic Treaty and Madrid Protocol environmental obligations.

6 Scalability & future scope

Isn't three stations too niche?	The polar station is the hardest case of a common problem — an island grid running on delivered fuel.
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Where else does it apply?	Himalayan border posts, island grids, offshore platforms, telecom towers, disaster-relief camps.
How hard is it to add a station?	A configuration change — station profiles are data, not code.
What breaks at larger scale?	Nothing structural; a bigger asset mix just means a slightly larger optimisation.
Would it help Maitri II?	More than Bharati — a renewable-heavy station has more intermittency and more decisions to make.
Could it size a new station's design?	Yes — the simulator can answer how much battery, turbine and fuel reserve to build.
What's next after the hackathon?	Real telemetry, a water and waste-heat extension, and a multi-station fleet view.
Where would you be in six months?	Validated on real data, with a hardware-in-the-loop rig and a sizing study for Maitri II.

7 Risks, safety & failure

What if the system makes a bad decision?	Life-safety loads are hard constraints — the solver cannot trade them away at any price.
What if the optimiser crashes?	A deterministic rule-based dispatcher takes over and always returns a safe schedule.
What if the whole PC dies?	The station reverts to today's manual operation — we're a layer on top, never a dependency.
Do you add a single point of failure?	No — removing our system returns the station exactly to its current practice.
What about cybersecurity?	It's offline and effectively air-gapped, which removes most of the attack surface.
Any data privacy concerns?	Minimal — we use crew headcounts, never individual data.
What if a generator fails in winter?	N-1 reserve is a hard constraint, so the schedule already survives losing the largest unit.
What's your biggest risk?	The data — every number is a labelled projection until NCPOR shares real telemetry.
What's the weakest part of the idea?	Same answer — say it before they do, and never claim you have no weakness.

8 Users & adoption

What does the operator actually see?	Days of autonomy, what to do in the next 24 hours, and why.
Why would an experienced engineer trust it?	They shouldn't at first — it advises, shows its reasoning, and can always be overridden.
Does it need training to use?	Minutes — it's one screen answering three questions in plain language.
What if they ignore its advice?	Then it costs nothing and the station is no worse off than today.
How do you handle resistance to change?	Advisory mode, visible reasoning, and being measurably right where the engineer can check it.
Are you replacing the station engineer?	No — we give them a decision they'd have made anyway, earlier and with the arithmetic done.

9 Team & process — fill these in tonight, don't improvise

These are the questions teams fluff. Agree the answers as a group before you walk in, and make sure every member can name one thing they personally own.

Introduce your team.	Names, branch, and one line each on what they own in this project.
Who did what?	Name a clear owner for simulation, optimisation, ML, frontend and the pitch.
You're EEE — why an AI project?	Because the hard part isn't the AI, it's the power systems; a CS team would get the dispatch wrong.
How did you divide the work?	By layer, with the optimiser started first because it's the highest-risk component.
What was your biggest disagreement?	Have a real one ready — it shows you actually worked together.

What went wrong during the build?	The electricity-only design shut the generators off and lit the boiler — that failure produced the whole idea.
What did you learn?	That the obvious optimisation was wrong, and why domain knowledge beat more modelling.
How did you validate your assumptions?	Against published Antarctic station energy data, not intuition.
Will you continue after SIH?	Say yes and mean it — and name the next concrete step.

10 Curveballs

Why should we select your team?	Because we understood why the obvious answer is wrong — and we can show the arithmetic.
Convince me in 30 seconds.	One ship a year; generators waste fuel at low load; you can't switch them off because they're the heating.
What would you do differently?	Start from the thermal coupling instead of discovering it halfway through.
What if we gave you unlimited budget?	It wouldn't change much — the bottleneck is data access, not money.
Isn't this over-engineered?	The alternative is a rule of thumb that silently wastes a month of fuel a year.
Why should NCPOR listen to students?	They shouldn't take our word — they should take the measurement, which is why we report against a baseline.
A figure you can't recall?	Say you don't know, then say where you'd get it and what it would change.
Do you have any questions for us?	Always yes — ask whether they'd help open a channel to NCPOR for real data.

11 Lines worth memorising

"Fuel isn't a cost line in Antarctica. It's the survival margin."

"Switching the generator off to save fuel turns the boiler on. Electricity and heat are coupled — optimise them separately and you get the wrong answer."

"We don't add a point of failure. Turn us off and the station is exactly where it is today."

"Mawson cut diesel by two-thirds and went from refuelling every year to every second year. Fuel efficiency converts directly into resupply resilience."

"This PS sits in the software category, but it's an electrical engineering problem wearing a software coat."